1 Tier 2 Student Guide - From Pointwise Uncertainty to a Test Envelope

1.1 The question Tier 2 adds

In Tier 1, you often asked: what is the uncertainty in this result at this data point? In Tier 2, ask: how does the uncertainty change when the test condition changes?

Examples include airspeed uncertainty versus freestream velocity, C_l or C_d uncertainty versus wind-tunnel velocity, base-rate uncertainty versus applied reaction-wheel torque, and supporting gyro-rate uncertainty versus angular rate.

The output is a *condition-dependent uncertainty envelope*: a curve showing the uncertainty of a derived quantity across a useful range of test conditions.

1.2 Keep three quantities separate

1. *Measured inputs:* quantities such as voltage, differential pressure, current, resistance, angular rate, or encoder angle.
2. *Derived output:* a quantity calculated from the inputs, such as power, airspeed, lift,

C_l, C_d, or predicted base rate.

1. *Test condition:* the quantity varied or used to organize the study, such as resistance, freestream velocity, applied torque, or angular rate.

The uncertainty in the derived output is not necessarily constant as the test condition changes.

1.3 From Taylor-series propagation to normalized uncertainty

Suppose the derived quantity is

$$y = f(x_1, x_2, \ldots, x_n).$$

For systematic standard uncertainties in the inputs, the first-order Taylor-series estimate is

$$b_{y,\mathrm{sys}}^2 \approx \sum_i \left(\frac{\partial y}{\partial x_i} b_{x_i,\mathrm{sys}} \right)^2,$$

When the input uncertainties are treated as independent. If a covariance matrix is supplied, add the appropriate covariance terms instead of assuming independence.

Here $b_{x_i,\mathrm{sys}}$ means the systematic standard uncertainty assigned to input x_i. It may combine accuracy, calibration, resolution, quantization, gain, offset, or other setup-level components.

1.3.1 Step 1: Divide by the nominal output magnitude

The absolute result $b_{y,\mathrm{sys}}$ has the units of y. Divide by the nominal output magnitude $|y|$ to obtain a dimensionless output uncertainty:

$$\frac{b_{y,\mathrm{sys}}}{|y|} \approx \frac{1}{|y|} \sqrt{\sum_i \left(\frac{\partial y}{\partial x_i} b_{x_i,\mathrm{sys}} \right)^2}.$$

For one input contribution, insert $|x_i|/|x_i| = 1$ so the input uncertainty is also written in normalized form:

$$\frac{\left|\frac{\partial y}{\partial x_i}\right| b_{x_i,\mathrm{sys}}}{|y|} = \left|\frac{x_i}{|y|}\frac{\partial y}{\partial x_i}\right| \frac{b_{x_i,\mathrm{sys}}}{|x_i|}.$$

This is the normalization step that is easy to skip when it is written too compactly. The factor $x_i/|x_i|$ is only a sign factor. Because uncertainty contributions are squared, the sign does not affect the combined standard uncertainty, but it does matter when interpreting whether an input increase makes the output increase or decrease.

1.3.2 Step 2: Define the dimensionless sensitivity coefficient

Define the signed dimensionless sensitivity coefficient as

$$S_i = \frac{x_i}{y}\frac{\partial y}{\partial x_i}.$$

It describes the local fractional response of the model:

$$\frac{\Delta y}{y} \approx S_i \frac{\Delta x_i}{x_i}.$$

The magnitude $|S_i|$ is used when ranking uncertainty contributions. For independent inputs, the normalized systematic standard uncertainty is therefore

$$\mathrm{NSU}_{y,\mathrm{sys}} = \frac{b_{y,\mathrm{sys}}}{|y|} \approx \sqrt{\sum_i \left[S_i \frac{b_{x_i,\mathrm{sys}}}{|x_i|}\right]^2}.$$

This normalized form assumes that the nominal input values x_i and the nominal output y are nonzero. Near $x_i = 0$ or $y = 0$, the relative form and the coefficients S_i are undefined or misleading. Use the absolute form $\mathrm{SSU}_{y,\mathrm{sys}}$ instead, and do not interpret or plot $\mathrm{NSU}_{y,\mathrm{sys}}$ through that zero.

For a particular input, define its normalized contribution to the output as

$$\mathrm{NSU}_{x_i \to y} = |S_i| \frac{b_{x_i,\mathrm{sys}}}{|x_i|}.$$

This separates two ideas:

- S_i is the sensitivity of the data-reduction equation.
- $b_{x_i,\mathrm{sys}}/|x_i|$ is the relative quality of the measurement at the current test condition.

1.3.3 Step 3: Scale the normalized result back to output units

The normalized curve is dimensionless, but a client also needs to know the expected uncertainty in the units of the reported result. Multiply by the nominal output magnitude:

$$\mathrm{SSU}_{y,\mathrm{sys}}(q) = |y(q)|\,\mathrm{NSU}_{y,\mathrm{sys}}(q) = b_{y,\mathrm{sys}}(q),$$

where q is the test condition. We will call $\mathrm{SSU}_{y,\mathrm{sys}}$ the *scaled systematic standard uncertainty*. It has the units of the output: m/s for airspeed, N for lift, m for deflection, kg·m^2 for inertia, or W for power.

The two plots answer different questions:

- $\mathrm{NSU}_{y,\mathrm{sys}}(q)$ asks, "What fraction of the result is uncertain?"
- $\mathrm{SSU}_{y,\mathrm{sys}}(q)$ asks, "How large is the uncertainty in physical units?"

1.4 What do you do with the sensitivity coefficients?

For each test condition:

1. Evaluate the nominal inputs and the nominal output from the data-reduction equation.
2. Calculate or estimate the systematic standard uncertainty of each input.
3. Calculate S_i for every input that appears in the equation.
4. Calculate each $\mathrm{NSU}_{x_i \to y}$ contribution.
5. Combine the contributions by root-sum-square to obtain $\mathrm{NSU}_{y,\mathrm{sys}}$.
6. Multiply by $|y|$ to obtain $\mathrm{SSU}_{y,\mathrm{sys}}$ in output units.
7. Repeat over the full test-condition range and plot both curves.

The low point in the scaled curve is not automatically the best test condition. A useful condition must also be physically valid, safe, within the instrument range, relevant to the client question, and supported by a model that still applies.

1.5 Example of the interpretation

Suppose a sensor has a fixed resolution uncertainty a_x and a proportional accuracy $g_x|x|$. If a_x is the full width of a rectangular resolution interval, a convenient standard-uncertainty model is

$$b_{x,\mathrm{sys}} = \sqrt{\left(\frac{a_x}{\sqrt{12}}\right)^2 + (g_x|x|)^2}.$$

Then

$$\frac{b_{x,\mathrm{sys}}}{|x|} = \sqrt{\left(\frac{a_x}{\sqrt{12}|x|}\right)^2 + g_x^2}.$$

In this example, g_x is already defined as a relative one-standard-deviation value, while a_x is the full width of a rectangular interval. At small signal magnitude, the fixed-resolution term dominates. At larger signal magnitude, the curve approaches a floor set by g_x; this one-input expression does not create a minimum. A bucket-shaped minimum appears when different inputs dominate at opposite ends of the test-condition range.

When a datasheet gives only a bare manufacturer statement such as $\pm h_x|x|$, the default simplified model is a bounded rectangular distribution with standard uncertainty $h_x|x|/\sqrt{3}$, unless the datasheet states a confidence level or a one-sigma interpretation. A DMM specification written as ±(% of reading + digits) is usually a worst-case additive bound; combining its components by root-sum-square is a modeling choice that should be stated, not an automatic consequence of the specification.

1.6 Monte Carlo connection

Monte Carlo uses the same measurement model without requiring every partial derivative. For each nominal test condition:

1. Set the centers of the input distributions to the nominal values.
2. Draw one plausible value for each uncertain input.
3. Evaluate the data-reduction equation.
4. Repeat many times.
5. Compute the output standard deviation and divide by the nominal output magnitude.
6. Repeat for the full test-condition range.

For a proportional Gaussian error and a bounded rectangular resolution error, one synthetic measurement can be represented as

$$x^* = x + g_x|x|Z + \frac{a_x}{2}U,$$

where Z is a standard normal draw and U is a uniform draw between -1 and 1. The standard deviation of the resulting output samples estimates $\mathrm{SSU}_{y,\mathrm{sys}}$; dividing by $|y|$ estimates $\mathrm{NSU}_{y,\mathrm{sys}}$.

Monte Carlo does not eliminate the need to define what each distribution means. Use a common draw for a setup-level calibration bias across a virtual experiment when that bias should remain correlated across samples; do not redraw that bias independently at every time sample.

The Taylor-series and Monte Carlo curves should agree when the uncertainties are small and the model is locally linear. Differences in the standard deviation can occur when the uncertainty is large relative to the model scale, the nominal output is near zero, an input correlation is omitted, or the model is nonsmooth. For a linear model, the Taylor root-sum-square standard deviation is exact for any input distributions with finite variance. Bounded distributions mainly change the shape of the output distribution and therefore affect percentile or coverage intervals, not the standard deviation itself.

1.7 Foundational references

The following sources provide the broader technical foundation for the methods used in this guide. They are references for going deeper, not a requirement to read every page.

1. *Coleman and Steele,* Experimentation, Validation, and Uncertainty Analysis for Engineers*, 4th ed.* Wiley, 2018. Chapters 1, 3, and 4 are especially relevant to uncertainty sources, Taylor-series propagation, Monte Carlo propagation, and experiment planning. This is the course's primary working reference. Wiley Online Library
2. *JCGM 100:2008,* Evaluation of measurement data - Guide to the expression of uncertainty in measurement *(GUM).* This is the international reference for evaluating and expressing measurement uncertainty. BIPM publication page
3. *Taylor and Kuyatt,* Guidelines for Evaluating and Expressing the Uncertainty of NIST Measurement Results*, NIST Technical Note 1297.* This practical guide develops standard uncertainty, Type A and Type B evaluations, combined standard uncertainty, and reporting conventions. NIST Technical Note 1297

4. *Possolo,* Simple Guide for Evaluating and Expressing the Uncertainty of NIST Measurement Results, *NIST Technical Note 1900.* This is a more recent NIST guide with worked examples and additional discussion of uncertainty models and Monte Carlo methods. NIST Technical Note 1900
5. *ASME PTC 19.1,* Test Uncertainty. This professional standard addresses uncertainty in test measurements and propagation to test results. It is a useful reference when the experiment is being treated as an engineering test rather than only a classroom exercise. ASME standard page

These sources use overlapping terminology but are not interchangeable documents. Coleman and Steele is the most accessible starting point for this course; the GUM and NIST documents provide the formal measurement-uncertainty foundation; and ASME PTC 19.1 provides a test-engineering context.

1.8 How to turn the curves into a test decision

For each region of the plot, identify the dominant input contribution, compare the output uncertainty with the client's acceptance criterion, check hardware and model limits, and decide whether an additional test point would be useful. Then propose a specific improvement, such as a different sensor range, improved resolution, a better calibration, a different sampling strategy, or an additional measurement. Quantify how the proposed improvement changes the two curves.

Use *relative standard uncertainty* only for a value at one condition, *condition-dependent uncertainty envelope* for the curve over a range of conditions, *scaled systematic standard uncertainty* for the output-unit curve, and *dimensionless sensitivity coefficient* for the model factor S_i. Avoid using r for relative uncertainty in this assignment because R is also used for electrical resistance.

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